SolaX Solar ESS EV Charging Solution

ENGLISH

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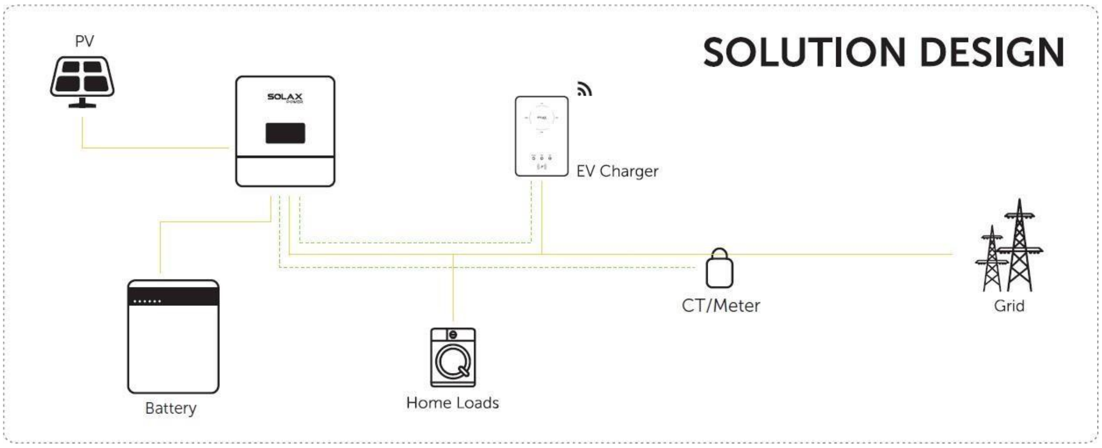
1 Solution Overview

Solax offers the Solar ESS EV Charging solution to make it possible to achieve intelligent energy management and charge your vehicle with green energy (energy from PV). This system includes PV panels, hybrid inverter, batteries, **EV charger (Home version)** and some necessary accessories. The schematic diagram is shown as below.



Solar ESS EV Charging System

2 Key Features

Intelligent charging modes

Thanks to the communication between the charger and inverter, the system has Green/Eco mode to use full-green/Semi-green energy to charge your vehicle.
In Green mode: The default level is 6A, in which the EV Charger will only start to charge if generating current reaches 6A, and it never take electricity from the grid. While there is another 3A level start to charge if generating current reaches 3A, capable to purchase other up to 3A from the grid. This work mode will spend all its effort to help clients use renewable energy and reduce the cost of buying electricity from the grid.

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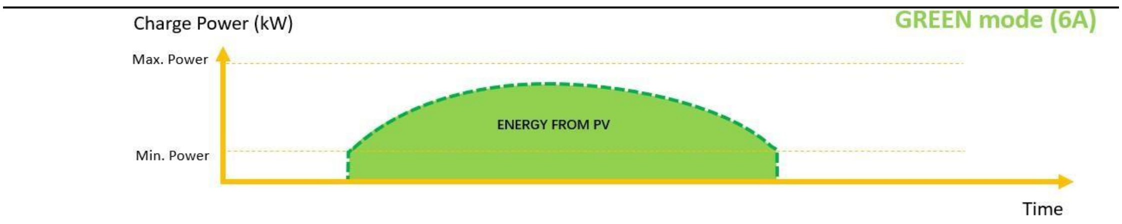
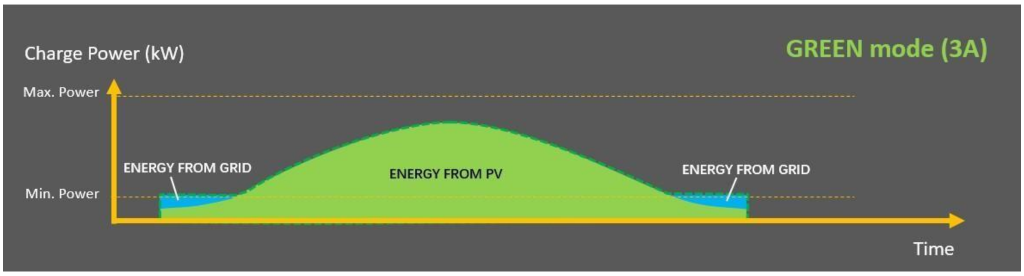
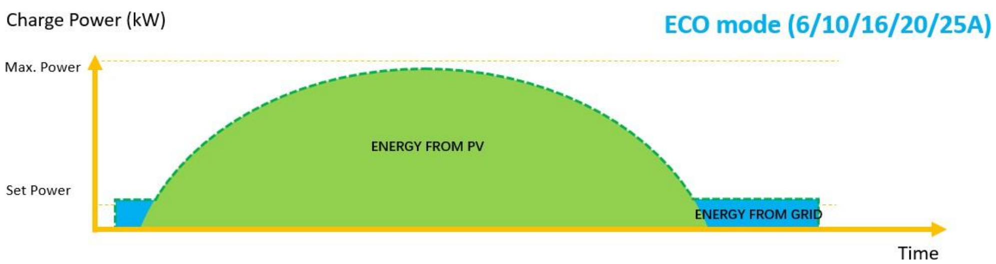


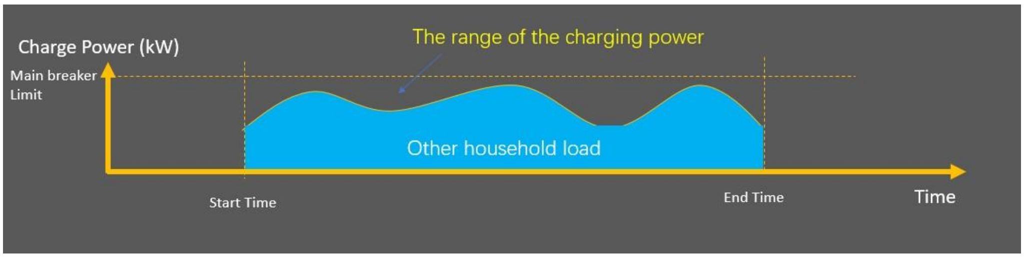


In Eco mode： it helps users to charge at a minimum power rate (set by user). If there is surplus energy form PV, it can exceed the minimum power rate to make full use the energy. If the PV energy is not enough, it will take the electricity from the grid to charge at the minimum power rate. For example, the users set the charging current 16A. If the current from the inverter is only 10A then the rest would be taken from the grid as 6A. If the current from the inverter(PV) is 18A, then the Smart EV Charger will output 18A.



Dynamic load balancing function

This function will make sure that the total power of house will not exceed the main grid capacity when using EV charger. It is realized by adjusting the charger power output automatically according to the value of the main grid current.



3 Communication Cable Connection

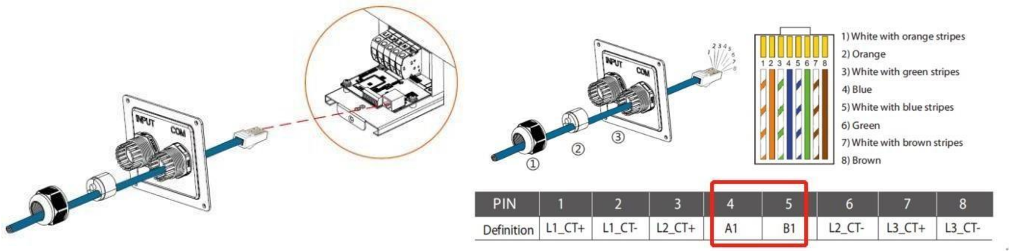
The EV charger need communicate with inverter via RS485, not all the inverters support this function. The inverter supported list are as below:

Model	X1-Hybrid G4	X3-Hybrid G4	X3-PRO G2	X3-MIC G2	X1-Mini G3/G4	X1-Boost G3/G4

And the inverter must connect a measuring device(meter/CT).

3.1 The COM ports of the EV charger

As the diagram shown the pin4/5(RS485) are used to communicate with the inverters. Please note the pin 1/2/3/6/7/8 are used to connect CTs in standalone system. You don't need use them if there is connection between EV charger and inverter.

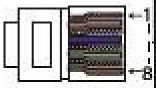


PIN	1	2	3	4	5	6	7	8
Definition	L1_CT+	L1_CT-	L2_CT+	A1	B1	L2_CT-	L3_CT+	L3_CT-

EV Charger COM interface

3.2 COM port of X1-Hybrid G4 & X3-Hybrid G4

Connect the EV charger to the COM port of the inverter via cat5 cable directly. Both side are RJ45 port and cable assignment is matching with each other.

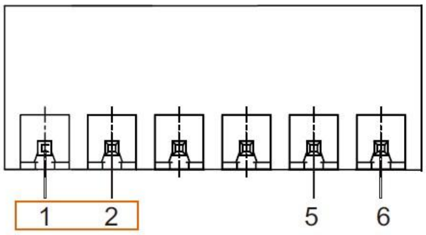


1	2	3	4	5	6	7	8
Drycontact_A(in)	Drycontact_B(in)	+13V	485A	485B	GND	Drycontact_A(out)	Drycontact_B(out)

Interface of COM of the hybrid G4 inverter

3.3 **COM port of X3-PRO G2**

Connect the pin4/5 of charger to the PIN 1/2 of the "RS485" port of the inverter. PIN 5/6 connect to the meter.

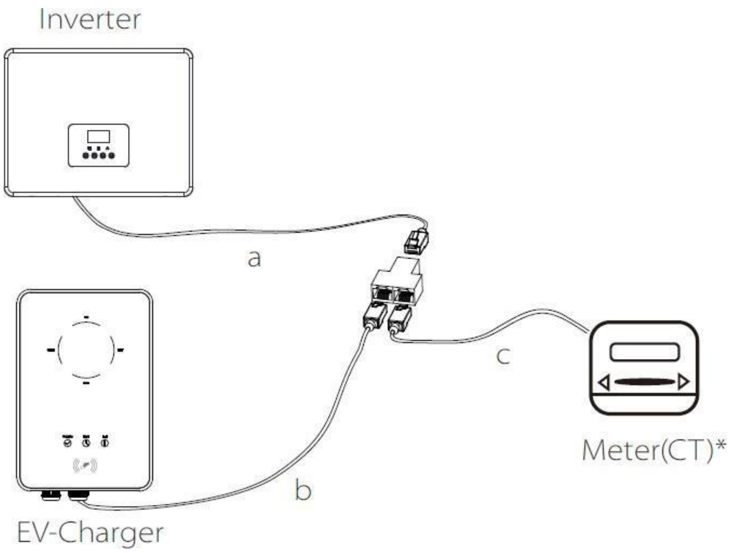


Pin	1	2	3	4	5	6
Definition	485A	485B	485A	485B	meter A	meter B

Interface of COM of the X3-PRO G2 inverter

3.4 **COM port of X3-MIC G2, X1-Mini G3/G4 & X1-Boost G3/G4**

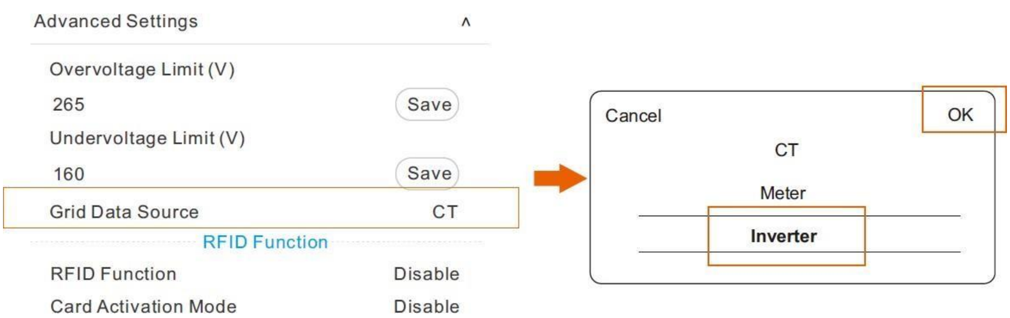
The RS485 port shares the same RJ45 port with Meter, so a splitter is needed to connect EV charger and meter to the inverter as the below diagram shown.
Please not the splitter should be placed in a waterproof place.



4 Settings

4.1 Setting on EV-Charger

When the EV-Charger is powered on, follow the instructions in the user manual to download the APP, complete the Wi-Fi connection and login.
Select "Advanced Settings" and set the "Grid Data Source" as "Inverter" as shown below and touch "OK" to confirm. The buzzer of the EV-Charger will beep after the setting is succeeded.



Set the Main Breaker limit as below if need enable the Dynamic load balancing function

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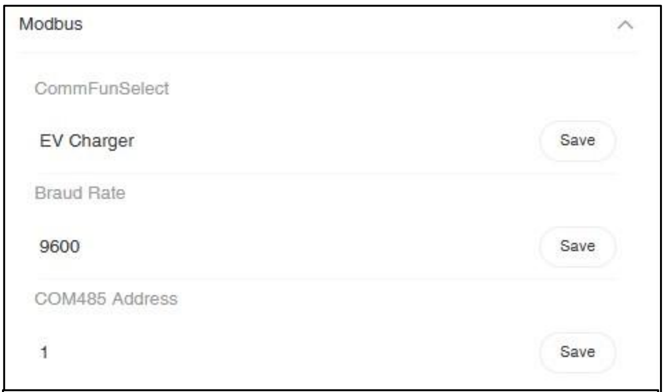
4.1 Settings on the Inverter

Comunication setting Go to Advanced -> Modbus
set as below:
CommFunSelect: EV charger
Braud Rate: 9600
EV Charger Address: 70



Battery charge EVC function
(Only for G4 hybrid inverter) Go to Advanced -> Battery charge
EVC.

Enable means storage battery allow EV charger to use battery’s energy.
Disable means storage battery doesn’t allow EV charger to use battery’s energy.

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Notes:

All the operations described in this guide can only be performed by qualified electricians with a good knowledge of the characteristics and maintenance of the smart EV-Charger and the inverter. Any operations are prohibited before reading this guide and the relevant manual carefully. All the settings for the smart EV- Charger and the inverter need to refer to the corresponding manual strictly. SolaX Power will not be responsible for any damages or harms caused by improper
setting.